

2016.0 RANGE ROVER (LG), 501-16

WIPERS AND WASHERS

DIAGNOSIS AND TESTING

PRINCIPLE OF OPERATION

The wiper and washer system is controlled by the body control module. All wiper functions for the front and rear wipers are controlled from a multi-function wash/wipe switch assembly located on the steering column

For a detailed description of the wipers and washers systems and operation, refer to the relevant description and operation section of the workshop manual. REFER to: [Wipers and Washers](#) (501-16 Wipers and Washers, Description and Operation).

INSPECTION AND VERIFICATION

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault and may also cause additional faults in the vehicle being checked and/or the donor vehicle

NOTES:

- If the control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the warranty policy and procedures manual (section B1.2), or determine if any prior approval programme is in operation, prior to the installation of a new module/component
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests

1. Verify the customer concern

2. Visually inspect for obvious signs of mechanical or electrical damage

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Windshield and rear window for damage or contamination e.g. road film or general residue deposits ▪ Wiper blades, arms and linkage for wear, security, damage and freedom of movement ▪ Windshield/rear window/headlamp washer fluid level ▪ Washer pipes and jets for leaks, restrictions and damage ▪ Wipers and washers control switch, damage and freedom of movement	▪ Battery condition and state of charge ▪ Fusible links ▪ Fuses ▪ Relays ▪ Electrical connections ▪ Front and rear wiper motors ▪ Wiper switch ▪ Washer pumps ▪ Rain/light sensor ▪ Heated front washer jets ▪ Ignition switch ▪ Light switch ▪ Ambient air temperature sensor ▪ Battery junction box

- Body control module
- ABS control module
- Automatic temperature control module
- Instrument cluster
- CAN circuits

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step
4. If the cause is not visually evident, verify the symptom and refer to the symptom chart, alternatively check for diagnostic trouble codes (DTCs) and refer to the DTC Index.

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
Wipers and washers inoperative	▪ Washers inoperative ▪ Wiper arm(s) incorrectly installed/aligned ▪ Steering column right multifunction switch internal failure ▪ Wipers and washers control switch circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Listen for washer motor operation. Check and top up washer fluid level. Check and rectify blocked washer circuit. Check for DTCs indicating a wiper/washer circuit fault. Rectify as necessary ▪ Check the installation/alignment of the wiper arms. Ensure motor/mechanism is not jammed or seized ▪ Check if the wipers and washers control switch retaining screws are tightened according to specifications. Inappropriately tightened screws might cause steering wheel module failure ▪ Refer to the electrical circuit diagrams and check the wipers and washers control switch circuit for short circuit to ground, short circuit to power, open circuit, high resistance
Wiper blade(s) drag/judder	▪ Contamination of the	Clean the windshield/rear window. Check for the correct installation and tension of the wiper

across the windshield/rear window	windshield/rear window ▪ Incorrectly installed wiper arm(s) ▪ Wiper arm(s) incorrectly aligned to the screen ▪ Wiper arm(s) spring tension inadequate	arm(s). Refer to the relevant section of the workshop manual. Rectify as necessary
Very slow operation of the wiper(s) across the windshield/rear window	▪ Low battery voltage ▪ Front wiper linkage seized or fouling ▪ Wiper circuit fault ▪ Wiper switch fault, high resistance	Check the battery condition and state of charge. Check the wiper linkage for fouling. Disconnect the motor from the linkage. Refer to the relevant section of the workshop manual. Check the linkage operation. Check for DTCs indicating a wiper circuit fault. Rectify as necessary
Wiper(s) inoperative		
Noisy operation of wiper(s)	▪ Wiper motor/linkage fault	Lift the wiper arm(s) from the windshield/rear window and recheck the noise level during the wiper sweep operation
Noisy operation of washers	▪ Washer motor(s) faulty ▪ Washer system blocked or partially blocked	Listen for washer motor operation. Check and top up washer fluid level. Check and rectify blocked washer circuit. Check the wiper/washer circuit for DTCs indicating a fault. Rectify as necessary
Washers do not operate	▪ Fluid level low ▪ Washer circuit blocked ▪ Washer circuit faulty	

DTC INDEX

For a list of diagnostic trouble codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00. REFER to: [Diagnostic Trouble Code Index - DTC: Body Control Module \(BCM\)](#) (100-00 General Information, Description and Operation).